3 Fisher Kernels

Let p_θ be a family of probability densities indexed by $\theta \in \mathbb{R}$ and let (X) be a set of points $\{x_i\}$. Then the Fisher kernel is given as

$$k(x_i, x_j) = \frac{\phi(x_i, \theta_0)\phi(x_i, \theta_0)}{I(\theta_0)}$$

where $\phi(x, \theta_0) = \frac{\partial}{\partial\theta} \ln p_\theta(x)|_{\theta=\theta_0}$ and

$$I(\theta) = E[\phi^2(X, \theta)] \text{ with } X \sim p_\theta$$

1. Show that $k(x_i, x_j)$ is positive definite.*first version forgot some α_i's in the sums

$$\sum_{ij} \alpha_i\alpha_j k(x_i, x_j) = \sum_{ij} \alpha_i\alpha_j \frac{\phi(x_i)\phi(x_j)}{I(\theta_0)} = \frac{\sum_i \alpha_i\phi(x_i)\sum_j \alpha_j\phi(x_j)}{I(\theta_0)} = \frac{\left(\sum_k \alpha_k\phi(x_k, \theta_0)\right)^2}{E(\phi^2(X, \theta))|_{\theta=\theta_0}} \geq 0$$

This follows becuase the numerator is ≥ 0 and the demoninator strictly greater than zero since otherwise this would imply that p_θ is independent of θ contrary to assumption.

2.Let $x \in \{0, 1\}$ and $X \sim$ Bernoulli(θ) with $0 < \theta < 1$, then $p_\theta = \theta^x(1-\theta)^{(1-x)}$. Evaluating $\phi(x, \theta)$ gives

$$\phi(x, \theta) = \frac{\partial}{\partial\theta}(x \ln\theta + \tfrac{\partial}{\partial\theta}(1-x)\ln(1-\theta)) = \frac{x}{\theta} + \frac{(1-x)}{(\theta-1)} = \frac{x(\theta - 1 + (1-x)\theta)}{\theta(\theta-1)} = \frac{(x-\theta)}{\theta(1-\theta)}$$

likewise for $I(\theta)$,

$$I(\theta_0) = E[\phi^2(X, \theta_0)] = E\left[\frac{(\theta - X)}{\theta(\theta-1)}\right]_{\theta=\theta_0} = \left.\frac{E(X-\theta)^2}{\theta^2(1-\theta)^2}\right|_{\theta=\theta_0} = \frac{\theta_0(1-\theta_0)}{\theta_0^2(1-\theta_0)^2} = \frac{1}{\theta_0(1-\theta_0)}$$

Therefore substituting both expressionss into $k(x_i, x_j)$ gives

$$k(x_i, x_j) = \frac{\frac{(x_i-\theta_0)}{\theta_0(1-\theta_0)}\frac{(x_j-\theta_0)}{\theta_0(1-\theta_0)}}{(1/\theta_0(1-\theta_0))} = \frac{(x_i-\theta_0)(x_j-\theta_0)}{\theta_0(1-\theta_0)}$$

3 Now Assume $x = (x_1, x_2)$ where $x_1, x_2 \in \{0, 1\}$ and are independent. Find $k(x^{(i)}, x^{(j)})$.

In this case $p_\theta = \theta^{x_1}(1-\theta)^{(1-x_1)}\theta^{x_2}(1-\theta)^{(1-x_2)} \Longrightarrow$

$$\phi(x, \theta) = \frac{\partial}{\partial\theta}\ln\left(\theta^{x_1}(1-\theta)^{(1-x_1)}\theta^{x_2}(1-\theta)^{(1-x_2)}\right)$$
$$I(\theta_0) = E[x_1 + x_2 - 2\theta_0]^2 = E[(x_1-\theta_0) + (x_2-\theta_0)]^2 =$$
$$E[(X_1-\theta_0)]^2 + E[(X_2-\theta_0)]^2 + 2E[(X_1-\theta_0)]E[(X_2-\theta_0)] =$$
$$\theta_0(1-\theta_0) + \theta_0(1-\theta_0) + 2(\theta_0-\theta_0)(\theta_0-\theta_0) = 2\theta_0(1-\theta_0)$$

This follows from the independence of x_1 and x_2.

Therefore $k(x^{(i)}, x^{(j)})$ is given by

$$\frac{\partial}{\partial\theta}\left(x_1^{(i)}\ln\theta + (1-x_1^{(i)})\ln(1-\theta) + x_1^{(i)}\ln\theta + (1-x_2^{(i)})\ln(1-\theta)\right) \times$$
$$\frac{\partial}{\partial\theta}\left(x_1^{(j)}\ln\theta + (1-x_1^{(j)})\ln(1-\theta) + (x_1^{(j)})\ln\theta + (1-x_2^{(j)})\ln(1-\theta)\right)/I(\theta_0) =$$

$$\frac{\left(x_1^{(i)} + x_2^{(i)} - 2\theta_0\right)\left(x_1^{(j)} + x_2^{(j)} - 2\theta_0\right)}{2\theta_0(1-\theta_0)}$$